

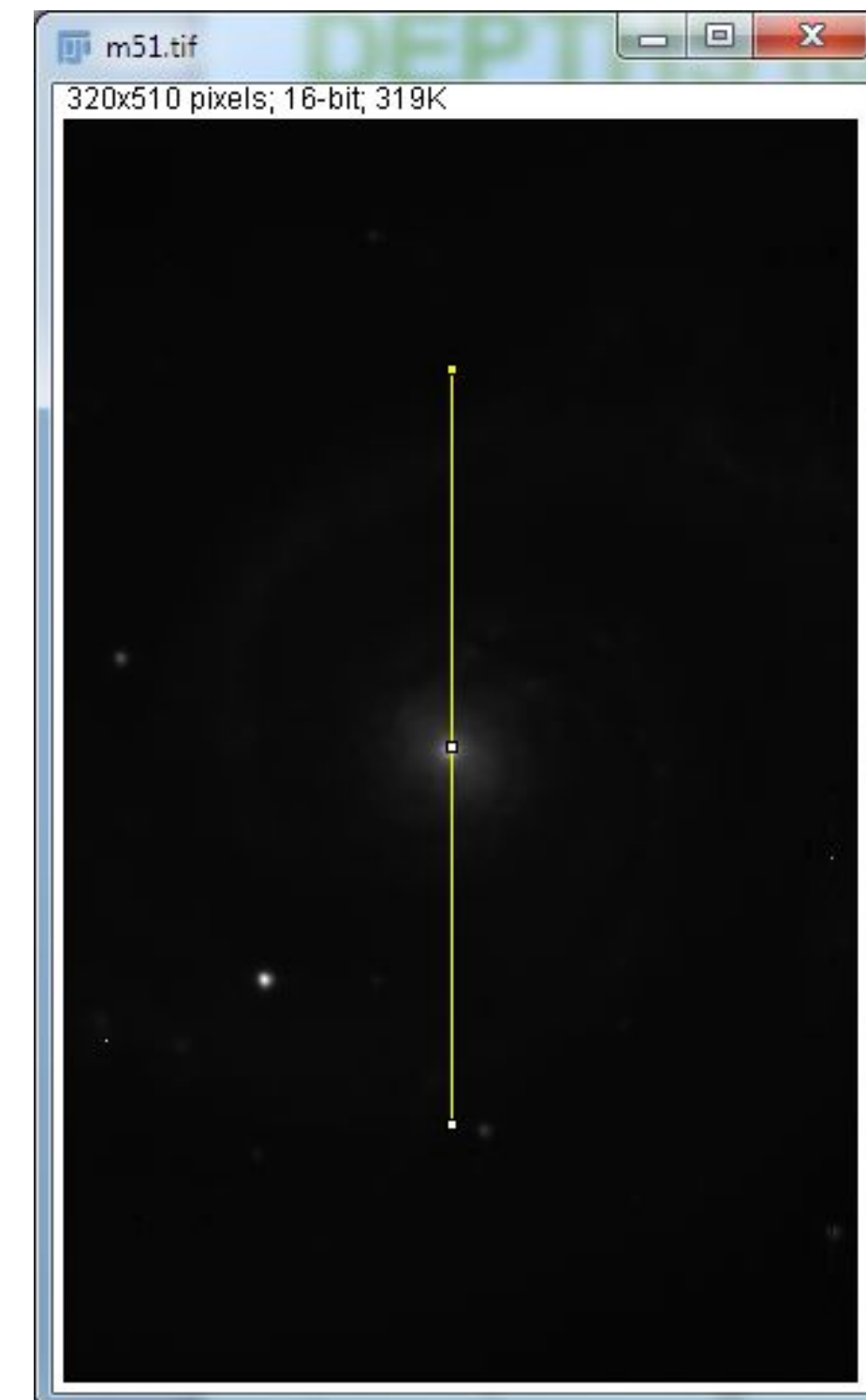
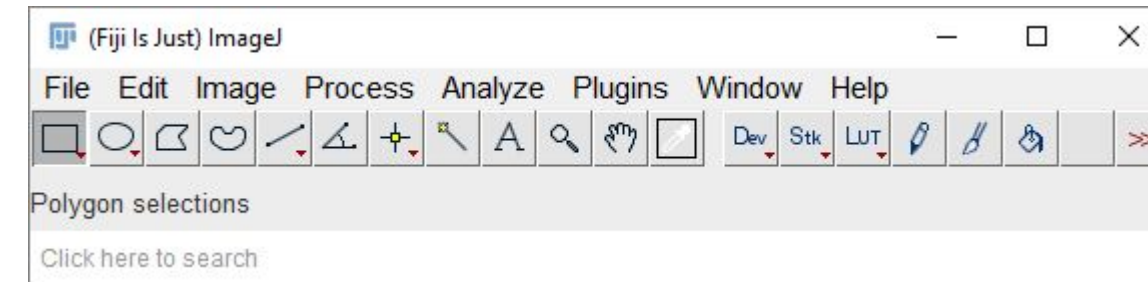
Image Bit-Depth

Image Processing & Analysis for Life Scientist

Olivier Burri, Romain Guiet & Arne Seitz

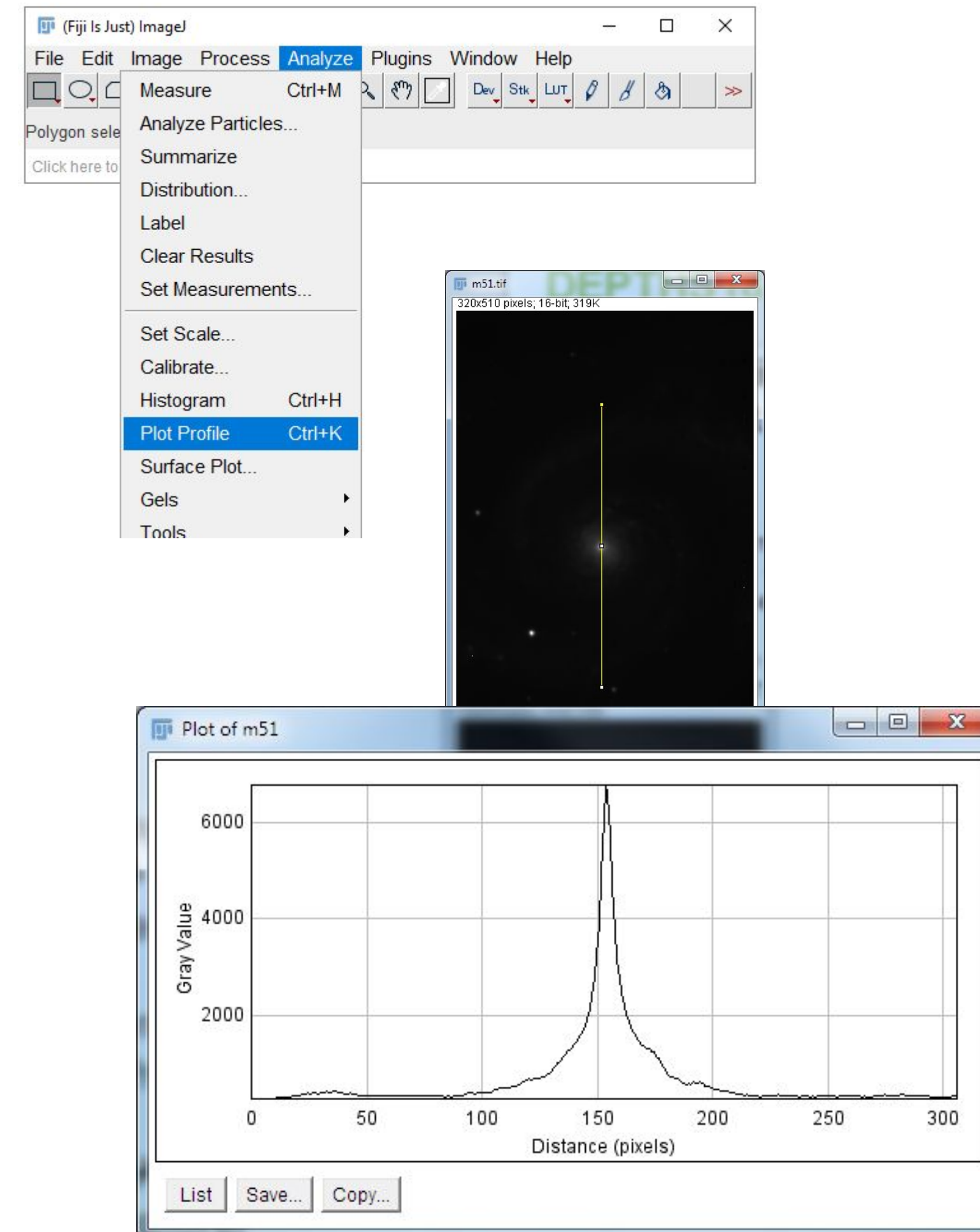
Open an Image and Make a Line

1. Open a 16-bit image from sample
File → Open Samples → M51 Galaxy
2. Choose “Line selection” tool
3. Draw a line



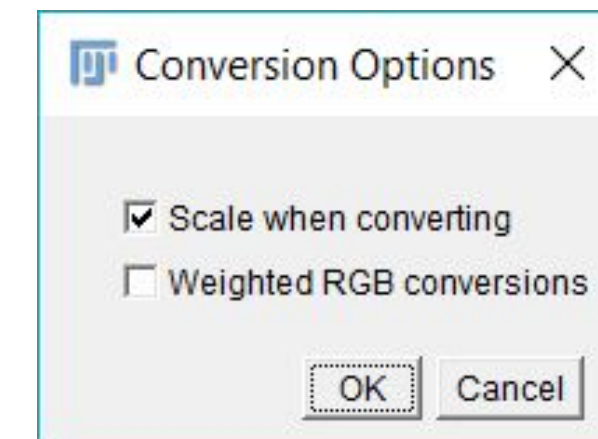
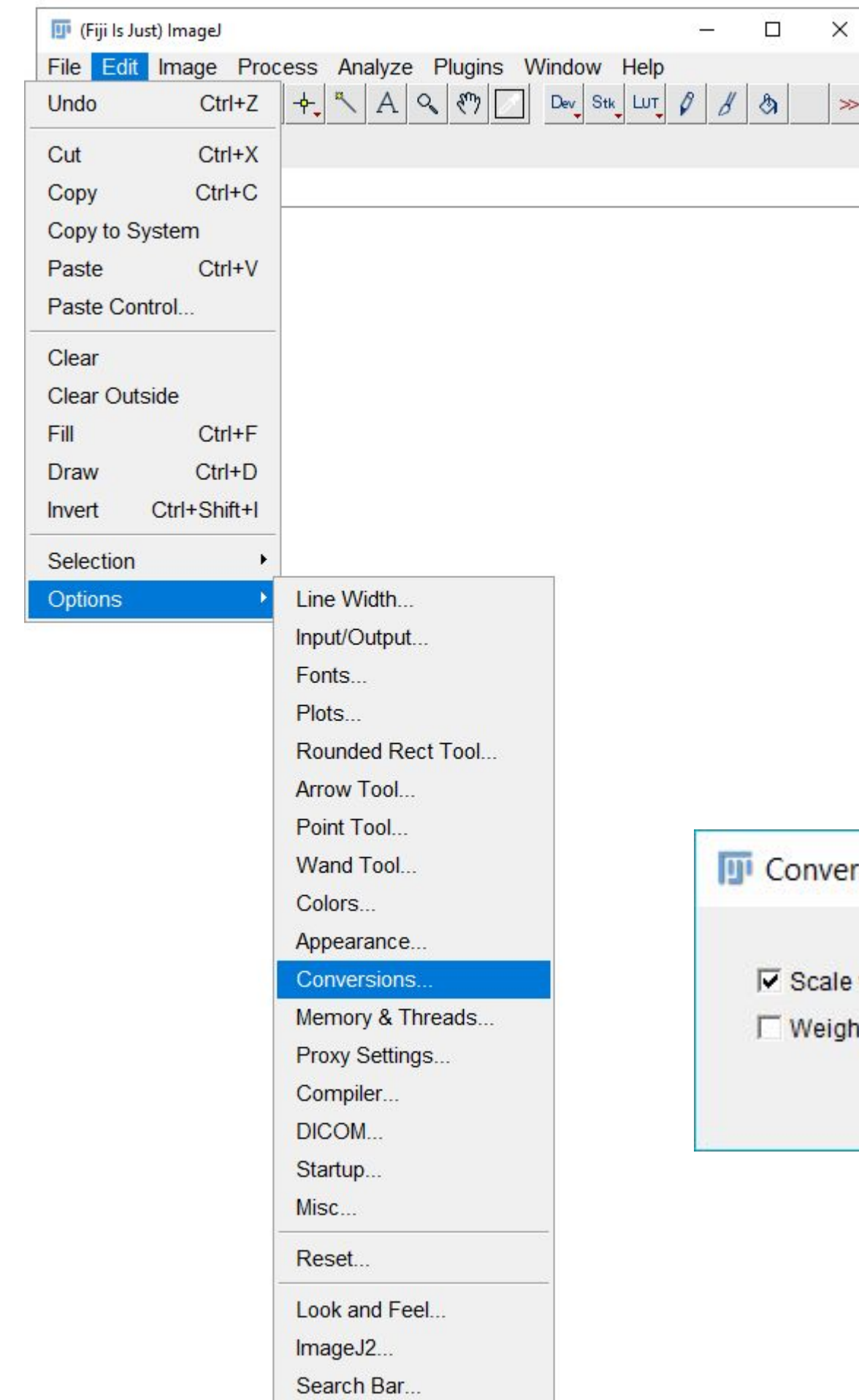
Get a Line Profile

1. Open a 16-bit image from sample
File → Open Samples → M51 Galaxy
2. Choose “Line selection” tool and draw a line
3. Draw a line
4. Look at the line profile using
Analyze → Plot Profile... (Shortcut K)



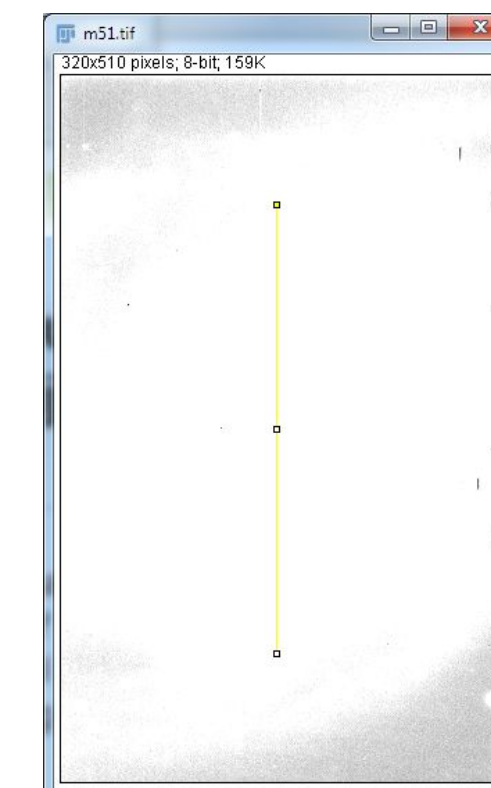
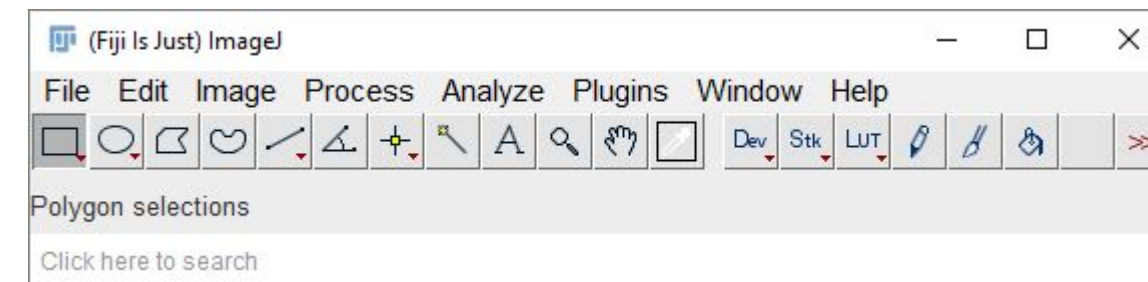
Convert to 8-bit and Make a new Profile

1. Turn bit-depth scaling off
Edit → Options → Conversions
“Scale when converting” : OFF
2. Convert the image to 8-bit
Image → Type → 8-bit



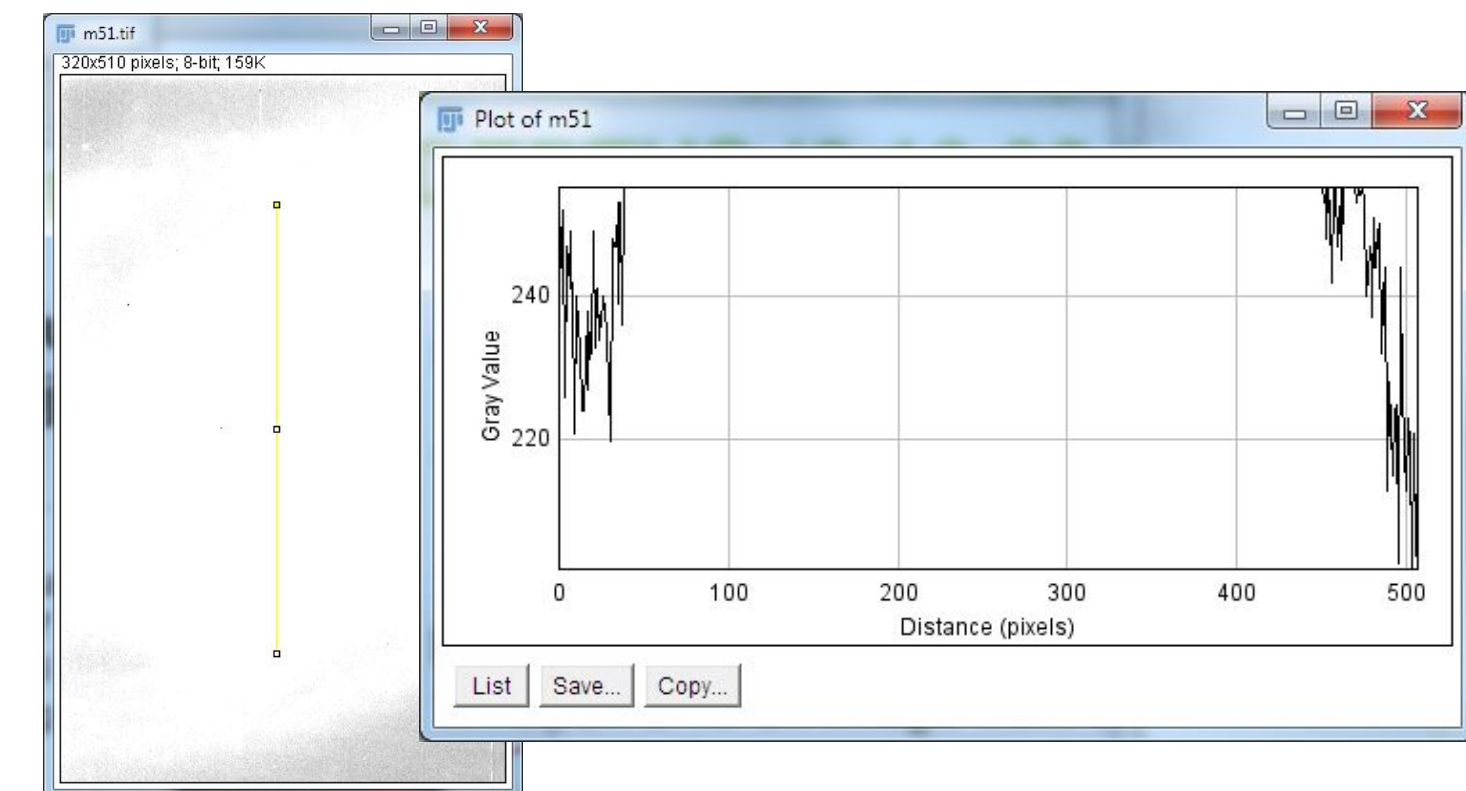
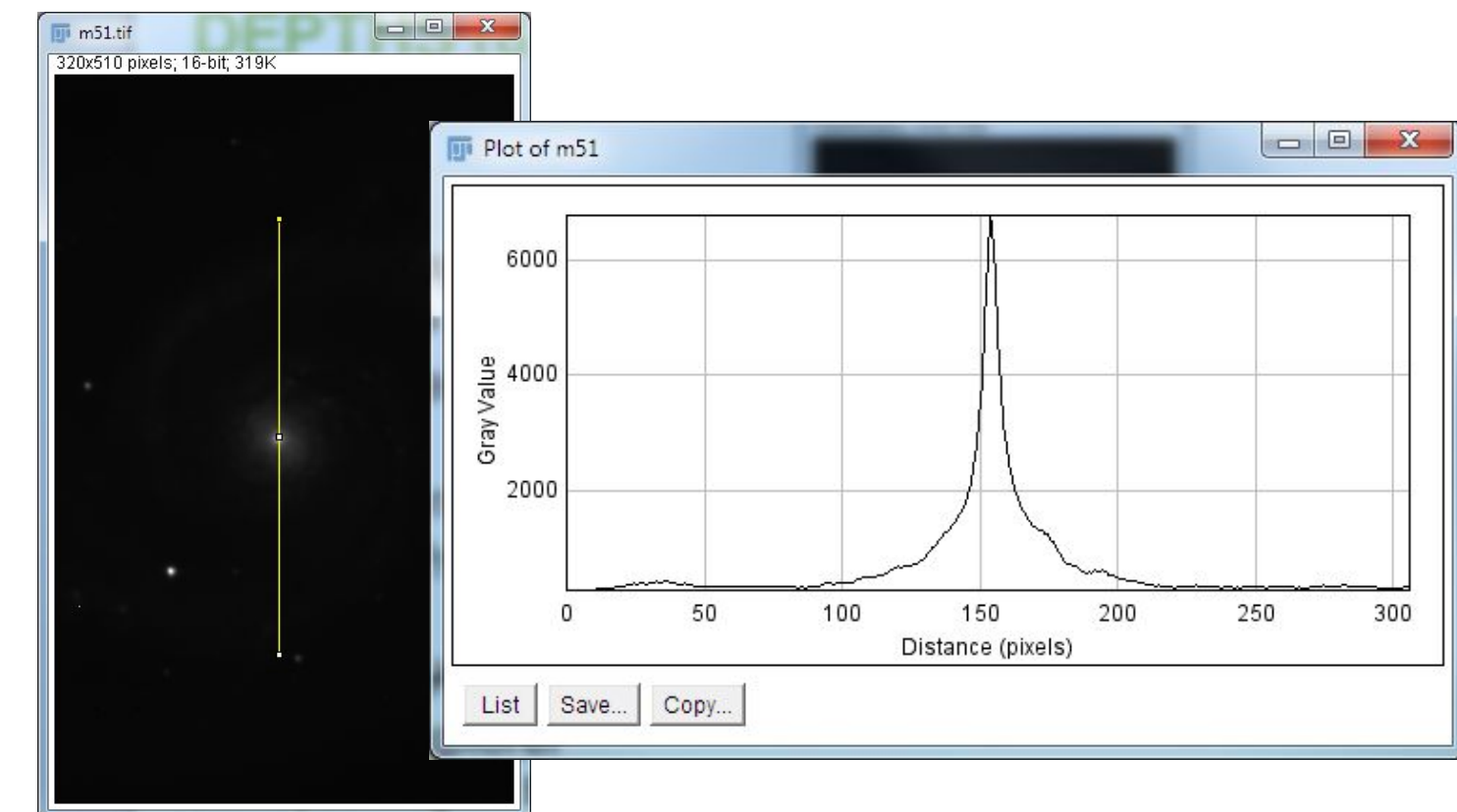
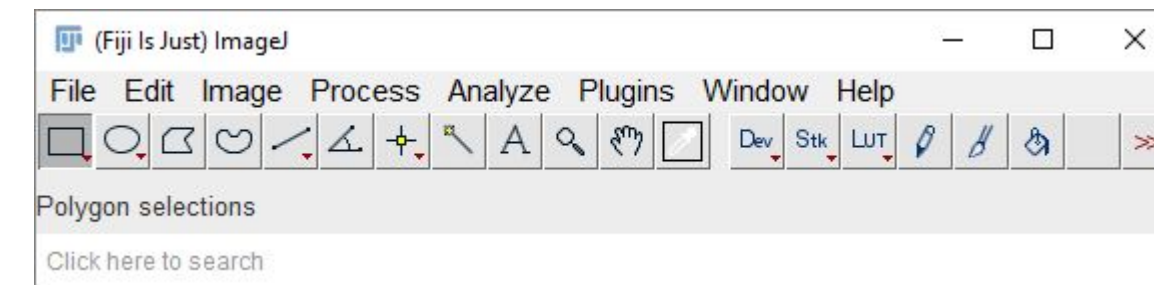
Convert to 8-bit and Make a new Profile

1. Turn bit-depth scaling off
Edit → **Options** → **Conversions**
“Scale when converting” : OFF
2. Convert the image to 8-bit
Image → **Type** → **8-bit**
3. Restore the Line
Edit → **Selection** → **Restore Selection**
4. Look at the new line profile.
Analyze → **Plot Profile...**



Convert to 8-bit and Make a new Profile

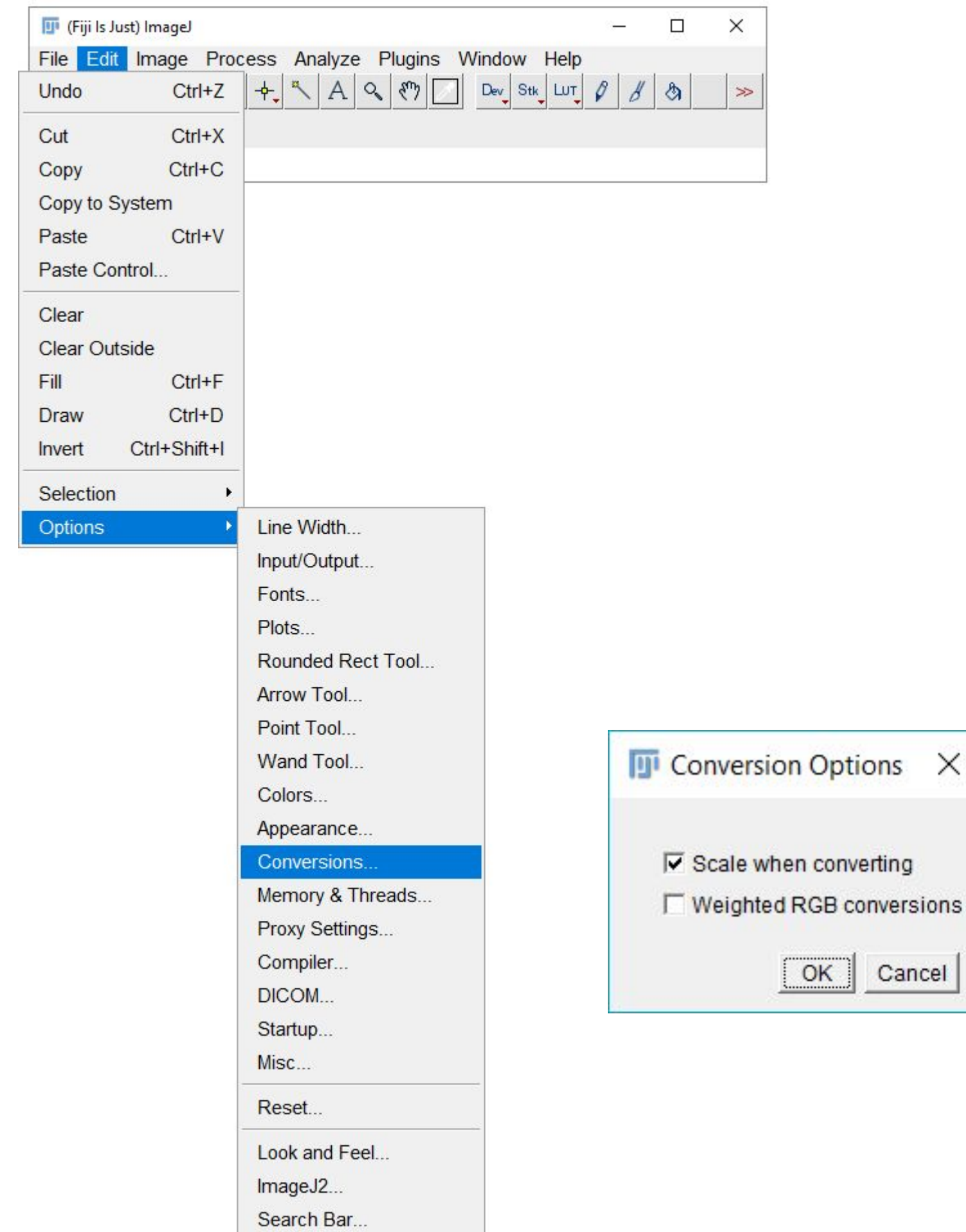
1. Turn bit-depth scaling off
Edit → **Options** → **Conversions**
“Scale when converting” : OFF
2. Convert the image to 8-bit
Image → **Type** → **8-bit**
3. Restore the Line
Edit → **Selection** → **Restore Selection**
4. Look at the new line profile.
Analyze → **Plot Profile...**



Proper Conversion

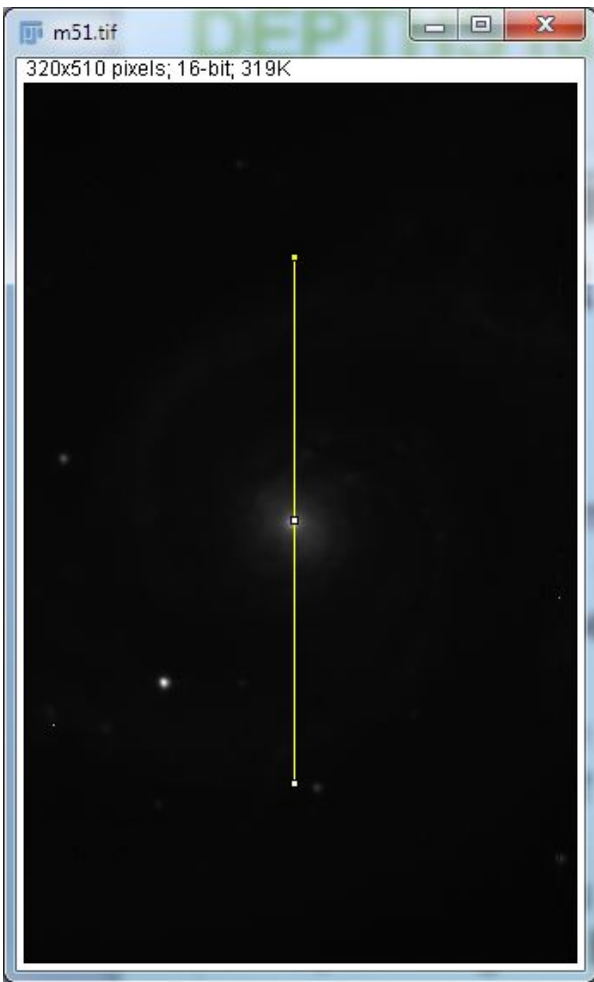
1. Re-open the image *M51 Galaxy*
2. Set “**Scale when converting**” to **ON**
Edit → **Options** → **Conversions**
“Scale when converting” : ON
3. Convert the image to 8-bit
4. Restore the Line
Edit → **Selection** → **Restore Selection**
5. Look at the new line profile.
Analyze → **Plot Profile...**

*Repeat, but change the **Brightness and Contrast** before the conversion.*

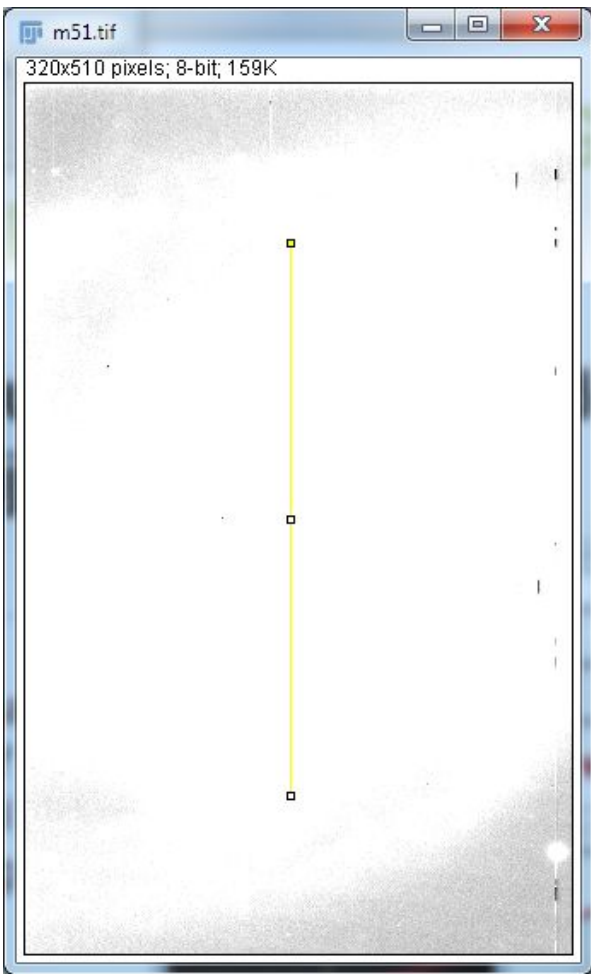


Take Care When Converting

16-bit



8-bit no scaling



8-bit scaling

